Three attack rationales for a uniform split-prime weighting: class-group relations, rational-isogeny lifting, modular-curve embedding

research/split_prime_ecdlp_20261107/rationales · 2026-11-08 · branch `claude/class-group-lifting-attacks-pw7y65`

Status: derived analysis of three candidate rationales, a native re-weighting of the lab's 21-curve census, and a PARI class-group computation for the smaller Frobenius fields; no algorithm claim. **Result class:** boundary study, follow-up to the parent split-prime study (§3 obstruction, §5 expander degree) and to the skipped-curve census (`nist_skip_criterion_20261025/strict_elkies_20261029`). `S` and end-to-end cost are unset; no leaderboard, benchmark or rho comparison is touched. Every measured number here is a single-producer value [pcc:unverified] except the 21 strict counts at 691, which match the strict study's table (third producer).

1. Question and answer

Request (user, 2026-11-08). Assess three stories under which a curve generator of 1997–98 would have counted split primes $\ell \le 691$ with uniform weight per prime, and whether any is an attack: (1) class-group relation attacks (each split $\ell = \mathfrak{l}\bar{\mathfrak{l}}$ gives a generator $[\mathfrak{l}]$ of $\mathrm{Cl}(\mathcal{O}_K)$ with a cheap Elkies realisation; Hafner–McCurley / Bisson–Sutherland relations need many small split primes of roughly uniform value); (2) lifting attacks needing rational isogenies (xedni 1998, anomalous 1997–98; a curve with many rational ℓ-isogenies is "closer" to reductions of curves on $X_0(\ell)$; weakness: rational ℓ-isogenies over $\mathbb{Q}$ exist only for $\ell \le 163$); (3) modular-curve embedding (split ℓ puts $j(E)$ on $X_0(\ell)(\mathbb{F}_p)$ with an explicit Frobenius eigenvalue on $E[\ell]$; a CRT-combining attack would cost proportionally to the number of usable ℓ; missing step: how per-ℓ data on $E[\ell]$ reach the subgroup of prime order $n \gg \ell$).

Answer. None of the three is an attack, and none predicts the weighting the census shows. (i) The class-group route gives only scalars on $E(\mathbb{F}_p)$ (parent theorem; prior form Theorem 15.1 of `RESEARCH_ISOGENY_FIBERS_ECDLP.md`), and its scale is wrong: a Hafner–McCurley factor base for a 194-bit discriminant reaches norms near 10^8 and the GRH generating bound $6\ln^2|\Delta|$ is near 10^5, so ten split primes below 691 change no class-group cost measurably; the one documented class-group defence in a standard (Brainpool, RFC 5639 §2.1 item 3) is a class-number floor $h > 10^7$ motivated by lifting and JMV reachability, not a split count. The class groups computed here (§4) do not separate published from skipped curves. (ii) The lifting route can use only Mazur's primes $\{2, 3, 5, 7, 11, 13, 17, 19, 37, 43, 67, 163\}$, at most two odd ones at once; the weight $[\ell \in M]$ separates at **no** cutoff, and at $B = 691$ the published P-256 curve has more Mazur-split primes (6) than its skipped sibling (5); the conductor claim is not a mechanism since isogenous curves over $\mathbb{Q}$ share their conductor. (iii) The modular-curve route CRT-combines Frobenius data mod ℓ into t, hence n, which is public; $E[\ell]$ and $E(\mathbb{F}_p)$ have coprime orders, so every per-ℓ datum is point-independent. Its weight $\ln \ell$ separates at 5 of 302 cutoffs against 4 for the uniform count: the census cannot tell them apart, and both are post-hoc fits. The census does locate the deficit: published-minus-skipped split fraction -0.107 at $\ell \le 50$, -0.082 at 51–163, -0.056 at 164–400, -0.025 at 401–691, -0.003 beyond 691 (descriptive two-proportion z $-1.8, -1.7, -1.5, -0.7, -0.2$): spread over $\ell \lesssim 400$, absent above the cutoff, no single band carrying it. Nothing here bears on the historical cause or reduces any ECDLP cost.

Scope. The lab's census of the four seed streams with both published and skipped curves (P-192: 3 + 2; P-224: 1 + 3; P-239: 3 + 5; P-256: 1 + 1) plus the published P-384 and P-521 singletons: 21 curves, p and n from `nist_skip_criterion_20261025/inputs.tsv`. Objects: $t = p + 1 - n$, $\Delta = t^2 - 4p = f^2 D_K$, $K = \mathbb{Q}(\sqrt{\Delta})$, $\chi(\ell) = (\Delta/\ell)$, $\mathrm{Cl}(\mathcal{O}_K)$.

Sources checked. KB unavailable (no `research-kb` server; `scripts/research-kb` finds no executable). Repository: parent README §3, §5, §6 and `ideas/idea-synthesist.md`; `followups/README.md` (P3 null); `strict_elkies_20261029/REPORT.md` and `results/at691.tsv`; `sea_replay_20261026/REPORT.md` (SEA coverage proxy separates in neither direction); `RESEARCH_ISOGENY_FIBERS_ECDLP.md` §15.1; `problems/split-prime-pattern-scalar-only.md`. Literature: Mazur's prime list checked against two expository sources (arXiv 2209.03153, 2206.08891); RFC 5639 §2.1 read directly; Hafner–McCurley 1989, Bisson–Sutherland 2011, Jacobson–Koblitz–Silverman–Stein–Teske 2000 (xedni presented at ECC '98, 14–16 September 1998) and the X9.62 timeline (developed 1998, approved January 1999) confirmed by web search; Kenku's composite list, Bach's bound, conductor invariance, Lenstra 1996, JMV 2005, Shoup 1997 and SEA from memory.

2. What each rationale needs, and the weight it implies

Let $M = \{3, 5, 7, 11, 13, 17, 19, 37, 43, 67, 163\}$ and $W_B(E) = \sum_{\ell \le B \text{ odd}, \chi(\ell)=+1} w(\ell)$.

Story 1, class-group relations. $\mathrm{Cl}(\mathcal{O})$ acts simply transitively on the class; a split ℓ gives $[\mathfrak{l}]$, $[\mathfrak{l}]^{-1}$ and two horizontal isogenies at $\tilde{O}(\ell)$ to $O(\ell^2)$ field operations each. Hafner–McCurley collect relations over all split primes of norm $\le B_{\mathrm{HM}} = L_{|\Delta|}(1/2, c)$, $c \approx 1/\sqrt{2}$; for $|\Delta| \approx 2^{194}$ this is $B_{\mathrm{HM}} \approx e^{18.2} \approx 7.6 \times 10^7$, a factor base of about 2.3×10^6 primes, so ten split primes below 691 are a relative change of order 10^{-5}. Bach's GRH generating bound is $6 \ln^2 |\Delta| \approx 1.1 \times 10^5$ (P-192) and 1.9×10^5 (P-256). Neither scale is 691; a class-group rationale with a cutoff near 691 would have to be structural (is $\mathrm{Cl}(\mathcal{O}_K)$ generated by the first few dozen split primes, is h smooth?), which §4 measures. Implied weight: $w = 1$ (one generator, one column per prime) or $w \approx 2/\ell$ (smoothness yield). DLP: every class acts through an isomorphism commuting with scalars and every endomorphism is $\alpha \bmod (\pi - 1)$ (parent theorem); knowing $\mathrm{Cl}(\mathcal{O}_K)$ completely adds no generic operation. Brainpool's $h > 10^7$ is a defence against lifting with End preserved ($[L : \mathbb{Q}] \ge h$) and JMV/HR reachability, not against relations.

Story 2, lifting with rational isogenies. A rational ℓ-isogeny of a lift $\tilde{E}/\mathbb{Q}$ reduces to one of E, so split or ramified ℓ is necessary; the anomalous attacks need $t = 1$ and no isogeny; xedni and Silverman–Suzuki lift points and need Mordell–Weil dependence. Over $\mathbb{Q}$ cyclic isogeny degrees lie in Kenku's list (primes in $M \cup \{2\}$; two distinct odd primes ≥ 5 only via degrees 15 and 21), so a split $\ell > 163$ is invisible to any lift. Isogenous curves share their conductor, so "extra isogeny structure lowers the conductor" is not a mechanism. Implied weight $[\ell \in M]$ or $[\ell \le 163]$; the complement $[\ell > 163]$ should carry nothing. DLP: the reduced isogeny is a horizontal edge, a scalar action; the lift's own content is the standing xedni obstruction.

Story 3, modular-curve embedding. $\chi(\ell) \ge 0$ iff Frobenius has an eigenvalue in $\mathbb{F}_\ell$ iff E has a rational ℓ-isogeny iff $(j(E), C) \in X_0(\ell)(\mathbb{F}_p)$; the Elkies kernel polynomial gives λ and $t \equiv \lambda + p/\lambda$; CRT gives t. That is SEA, and every curve sits on $X_0(\ell)$ for about half of all ℓ. Implied weight $\ln \ell$ (information); the SEA-cost variant (product of usable ℓ past the Hasse width) was replayed by the lab and separates in neither direction (published P-256 191 vs skipped 211; published P-224 257 vs skipped 139, 179, 233). DLP: the data CRT combines are functions of (E, ℓ); $E[\ell]$ and $E(\mathbb{F}_p)$ have coprime orders, so no non-zero equivariant $\mathbb{Z}$-linear map exists between them and every ℓ-pairing with a point of order n is 1; what is reconstructed is n.

A fourth reading (speculation). A generator fearing transport of the instance to a weaker isogenous curve (Frey's Weil-descent talk at the same ECC '98; Galbraith 1999; GHS 2002 for extension fields) would minimise cheap rational isogenies: one edge pair per split ℓ up to an affordable Elkies degree, i.e. the strict count with a hard cutoff. Indistinguishable from story 1 in the census; it predicts no dependence on h. On prime fields the moved-to curve is equally hard (JMV, parent theorem). No source found.

3. Census re-weighting (measured, native)

Method. `tools/weightings.rs` (dependency-free Rust) computes $\chi(\ell)$ for every odd prime $\ell \le 2000$ (302 cutoffs, the strict study's family) for the 21 curves, forms W_B for eight weights, and records the cutoffs at which every published W_B is below every skipped W_B in each paired field ("lower"), the reverse ("upper"), and the margins at 691. Control: the 21 strict counts at 691 equal `at691.tsv` row for row.

$w(\ell)$	rationale	lower-separating cutoffs (of 302)	upper	margins at 691, published max → skipped min, P-192 / 224 / 239 / 256
1	story 1 generation; §2 mobility	**4**: 691, 701, 709, 769	0	60→61 / 50→53 / 57→61 / 57→58
$\ln \ell$	story 3 information	**5**: 37, 151, 173, 179, 601	0	334.0→331.5 × / 281.9→281.5 × / 316.4→329.0 / 306.4→315.9
$1/\ell$	story 1 smoothness	0	0	0.584→0.572 × / 0.309→0.469 / 0.441→0.490 / 0.586→0.526 ×
$1/\sqrt{\ell}$	intermediate	0	0	4.69→4.72 / 3.33→4.26 / 4.26→4.69 / 4.65→4.50 ×

$[\ell \in M]$	story 2 (Mazur set)	**0**	0	4⟶4 ✕ / 2⟶3 / 4⟶4 ✕ / 6⟶5 ✕
$[\ell \leq 163]$	story 2 (bound)	0	0	16⟶17 / 11⟶12 / 17⟶18 / 18⟶16 ✕
$[\ell > 163]$	story 2′s blind range	0	0	47⟶43 ✕ / 39⟶33 ✕ / 42⟶42 ✕ / 39⟶42
1 on $\chi \geq 0$	inclusive	0	0	63⟶61 ✕ / 53⟶57 / 59⟶65 / 59⟶60

The uniform row reproduces the strict study's four cutoffs (positive control). The Mazur weight fails in three of four fields at 691 and at every cutoff; the published P-256 curve has six Mazur-split odd primes (11, 13, 17, 37, 43, 163) against five for its skipped sibling. $[\ell > 163]$ alone fails everywhere: the separation needs both ranges, which a uniform count supplies. $\ln \ell$ separates at five cutoffs, none of them 691; with 302 correlated post-hoc cutoffs, four against five favours neither weighting.

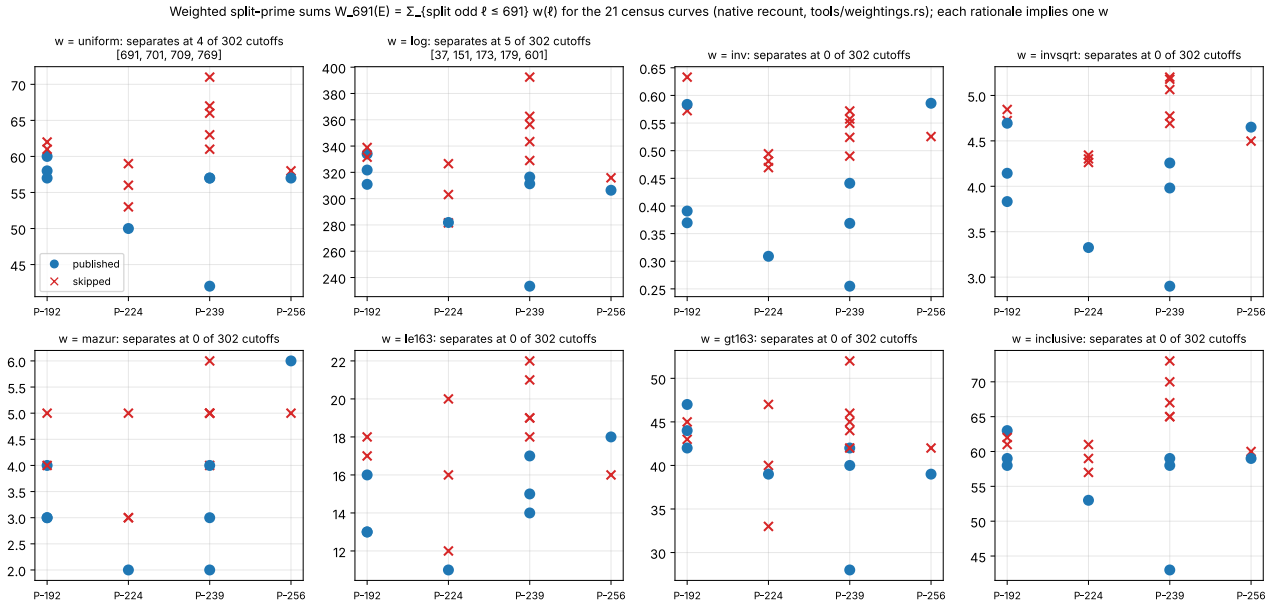


Figure 1: $W_{691}(E)$ per field for each weight; circles published, crosses skipped; titles give the number of separating cutoffs of 302. Source `figures/plot.py`, data `data/weightings.json` (native, `tools/weightings.rs`).

Where the deficit sits. Pooled over the four paired fields (8 published, 11 skipped curves):

band of odd ℓ	published split / cells	skipped split / cells	difference	two-proportion z (descriptive)
3–50	36 / 112 = 0.321	66 / 154 = 0.429	−0.107	−1.77
51–163	81 / 184 = 0.440	132 / 253 = 0.522	−0.082	−1.68
164–400	137 / 320 = 0.428	213 / 440 = 0.484	−0.056	−1.53
401–691	184 / 376 = 0.489	266 / 517 = 0.514	−0.025	−0.74
692–2000	702 / 1424 = 0.493	972 / 1958 = 0.496	−0.003	−0.20

The z values treat cells as independent trials although the cells of one curve share its discriminant: a yardstick, not a calibrated test. The deficit is spread over $\ell \lesssim 400$, weak per band, and absent above 691. A Mazur-only rule would leave 164–400 untouched; an information rule would put most of the deficit in the upper bands. The profile is compatible with a uniform count under a hard cutoff and with a weight decreasing slowly in ℓ; 19 curves cannot choose.

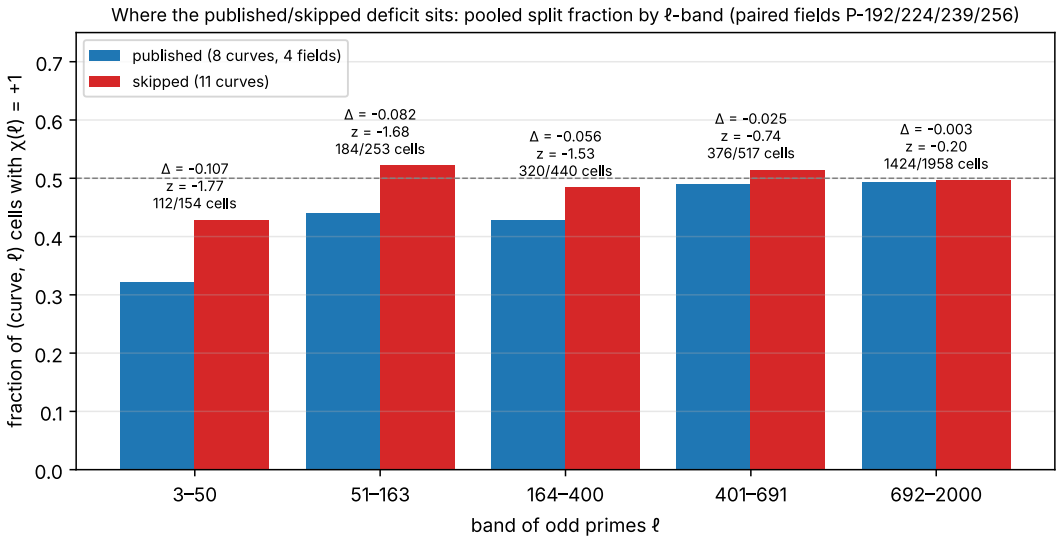


Figure 2: Pooled split fraction by ℓ-band, published against skipped, with difference, descriptive z and cell counts.

4. Class groups of the Frobenius fields (measured, PARI/GP 2.15.4)

The class number is the split pattern, to first order (derived). $h(\mathcal{O}_K) = (w/2\pi)\sqrt{|D_K|}L(1,\chi)$ with $L(1,\chi) = \prod_\ell (1-\chi(\ell)/\ell)^{-1}$, so $\log h = \frac{1}{2}\log|D_K| + \sum_\ell \chi(\ell)/\ell + O(1)$: a deficit of split primes **is** a class-number deficit. A defender who wanted a large class group (Brainpool's $h > 10^7$; the reading of "smooth, well-generated Cl is dangerous" as "small Cl is dangerous") would have selected for **more** split primes, the opposite of the observed direction. The observed direction is what a defender worried about availability of cheap edges or relations (§2) would pick, and §2 showed 691 is not a scale at which either matters.

Exact facts for all 21 curves (`tools/classgroup.gp`, stage `facts`: full factorisation of Δ, fundamental discriminant, conductor, partial Euler product for $L(1,\chi)$ over $\ell \le 10^6$; `data/cl/facts.jsonl`; 3 minutes, longest factorisation 360 s for P-384). The conductor is exact. Eighteen Frobenius orders $\mathbb{Z}[\pi]$ are maximal; the published P-224 has $f = 3$, prime239v2 (p239-2320, published) and the skipped p239-2682 have $f = 5$, so for those three $\mathrm{End}(E) \in \{\mathbb{Z}[\pi], \mathcal{O}_K\}$ is undetermined here. The h estimate is $(w/2\pi)\sqrt{|D_K|}$ times the partial product (heuristic tail): a size, not a class number.

label	role	bits	f	largest prime of Δ (bits)	$L(1,\chi)$ partial, $\ell \le 10^6$	$\log_2 h$ estimate
p192-139 (P-192)	published	192	1	184	0.411	94.0
p192-2198	skipped	192	1	157	0.498	93.6
p192-5228 (prime192v3)	published	192	1	84	0.352	93.8
p192-5405	skipped	192	1	119	0.434	94.0
p192-6013 (prime192v2)	published	192	1	179	0.529	93.6
p224-777	skipped	224	1	97	0.607	110.5
p224-1485	skipped	224	1	211	0.495	110.3
p224-5736	skipped	224	1	112	0.540	110.4
p224-7026 (P-224)	published	224	**3**	149	0.359	108.1
p239-36	skipped	239	1	205	0.436	117.1
p239-1562	skipped	239	1	160	0.590	118.1
p239-2185 (prime239v1)	published	239	1	217	0.246	116.8
p239-2320 (prime239v2)	published	239	**5**	132	0.476	115.2
p239-2491	skipped	239	1	177	0.415	117.6
p239-2682	skipped	239	**5**	81	0.558	115.7
p239-2863	skipped	239	1	149	0.505	117.7
p239-3048 (prime239v3)	published	239	1	239	0.410	117.5
p256-876	skipped	256	1	134	0.530	126.4
p256-1842 (P-256)	published	256	1	136	0.723	126.9

p384-2688 (P-384)	published	384	1	265	0.276	189.5
p521-1136 (P-521)	published	521	1	277	0.579	259.0

Does class-number size separate the roles? Per field, $L(1,\chi)$ partial (the factor carrying the split pattern): P-192 published 0.411, 0.352, 0.529 vs skipped 0.498, 0.434 (ratio of means 0.92; not separated, 0.529 > 0.434); P-224 0.359 vs 0.607, 0.495, 0.540 (0.66; published below all skipped); P-239 0.246, 0.476, 0.410 vs 0.436, 0.590, 0.415, 0.558, 0.505 (0.75; not separated, 0.476 > 0.415); P-256 0.723 vs 0.530 (1.36; **reversed**: the published curve has the larger class group). Class-number size separates in neither direction. The strict count separates where $L(1,\chi)$ does not because $L(1,\chi)$ weights ℓ by $1/\ell$ (the `inv` row of §3, zero separating cutoffs) and is dominated by $\ell = 3, 5, 7$, while the deficit is spread to $\ell \approx 400$. Whatever selected the published curves was not selecting on the class number.

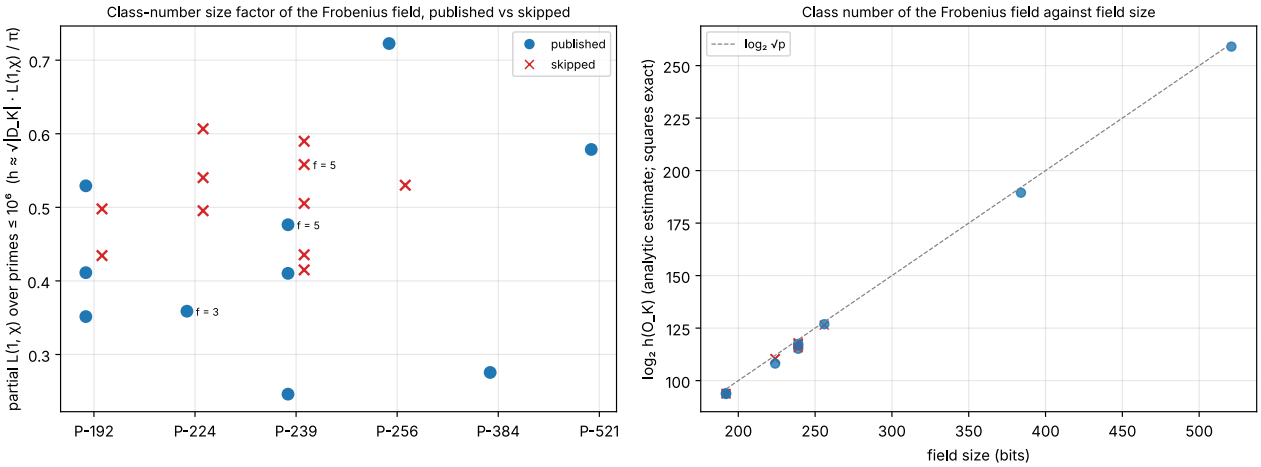


Figure 3: Left: partial $L(1,\chi)$ per field, published against skipped (annotated where $f > 1$). Right: $\log_2$ of the class-number estimate against field size; squares mark exact values where `quadclassunit` has finished. Data `data/cl/facts.jsonl`, `data/cl/cl_*.jsonl`.

Exact class groups and generation by the small split primes (P-192 field; pending at the time of writing). `quadclassunit` on this PARI build took 27 s for a 40-digit discriminant and did not finish a 45-digit one in 70 s or the 59-digit P-192 discriminant in 25 minutes. Exact $h(\mathcal{O}_K)$, its factorisation and the orders of the prime forms above the split ℓ (stage `orders`: index of the subgroup generated by split $\ell \le B$ for $B \in \{50, 163, 691, 2000\}$ as $h/\ \mathrm{lcm}$ of orders, exact when Cl is cyclic; the Mazur subset; the smallest ℓ whose form has full order) are running for the four other P-192 curves and are not attempted for the larger fields (P-224 to P-256 would be days). Whatever they return bears only on the structural reading of story 1 and cannot reverse §3 or the direction argument above; the GRH bound already gives generation by primes below $6\ln^2|\Delta| \approx 1.1 \times 10^5$, and the heuristic expectation is generation by the first handful of split primes for every curve. This paragraph is replaced by the results when they land.

5. Where each rationale meets the obstruction (derived)

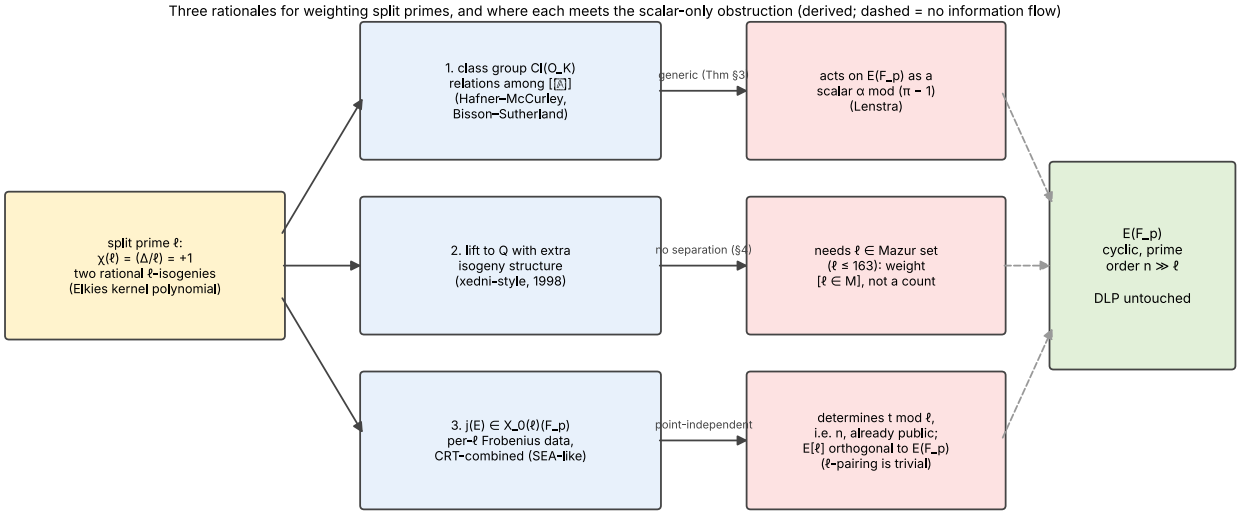


Figure 4: The three rationales and the point at which each loses contact with $E(\mathbb{F}_p)$. Derived; dashed arrows carry no information about the discrete logarithm.

rationale	object it reads	how it reaches $E(\mathbb{F}_p)$	status
1 class-group relations	$\mathrm{Cl}(\mathcal{O}_K)$, $[\mathfrak{l}]$ for split ℓ, relation lattice	isomorphisms φ with $\varphi([k]P) = [k]\varphi(P)$; endomorphisms as $\alpha \bmod (\pi - 1)$	scalar-only (parent §3 (1), (2)); knowing $\mathrm{Cl}(\mathcal{O}_K)$ adds no generic operation
2 lifting with isogenies	$\tilde{E}/\mathbb{Q}$ with a rational ℓ-isogeny, $\ell \in M$	reduction is a horizontal edge of E	scalar-only; the lift's DLP content is the standing xedni obstruction
3 modular-curve embedding	$(j(E), C) \in X_0(\ell)(\mathbb{F}_p)$, $\lambda \in \mathbb{F}_\ell$	CRT over ℓ gives t, hence n	point-independent (parent §3 (3)); reconstructs a public parameter
4 mobility (speculative)	horizontal degree $2 \cdot \#\{\text{split } \ell \le B\}$	transport cost of JMV walks	generic-model effect only (parent §5)

Nothing here exhibits the non-scalar map that `problems/split-prime-pattern-scalar-only.md` asks for; its status is unchanged.

6. Status of every statement

statement	status
separating-cutoff counts and margins, eight weights, 21 curves	measured natively; uniform row equals the strict study's four cutoffs; strict counts equal `at691.tsv` (third producer) [pcc:unverified]
band profile of the deficit	measured natively; z descriptive [pcc:unverified]
class numbers, structures, conductors, generation indices (§4)	PARI/GP under GRH; no certification; single producer [pcc:unverified]
scale arguments for story 1	derived from L-notation and Bach's bound (the latter from memory)
Mazur prime list	checked against two expository sources; Kenku's composite list from memory
conductor invariance under isogeny	standard, from memory
RFC 5639 requirement and motive	read directly
obstruction statements	derived in the parent study; Lean obligations in aburan28/crypto-lean#3

fourth reading	speculation, labelled; no source found
historical cause	the strict study's conclusion stands: no authenticated acceptance trace

Next steps. (1) Pre-registered out-of-sample prediction: under a uniform strict count with a hard cutoff in [691, 769], any further published curve of these streams has W_{691} at or below its field's published envelope (60, 50, 57, 57) and a larger skipped set has a flat deficit profile up to the cutoff; no such curves exist, and Brainpool's documented procedure is not a test of these streams. (2) P-384 and P-521 class groups need a dedicated run; nothing in §4 suggests it would change the reading. (3) No new Lean obligation.